

Lecture 19 Notes

Recap: Norms and the p -Norm Family

The lecture continues the normed-spaces topic. Recap from last time: if you are constrained to a rectangular grid (no diagonal moves), the distance to a point is the **absolute sum** of its components — the **one-norm**, also called the **taxicab** or **Manhattan** norm. It is one alternative to the Euclidean norm and, as the homework will show, a very powerful norm for **sparsity** applications.

Norm Axioms (Review)

A **norm** maps the set of vectors to the reals and must satisfy:

1. **Non-negativity:** $\|x\| \geq 0$ (it measures “bigness”).
2. **Definiteness:** $\|x\| = 0 \iff x = 0$ (only the origin has zero norm).
3. **Homogeneity:** $\|\alpha x\| = |\alpha| \|x\|$ (scaling scales the norm by the **absolute value** of the scalar).
4. **Triangle inequality:** $\|x + y\| \leq \|x\| + \|y\|$ (the third side of a triangle is bounded by the sum of the other two).

The Standing (Euclidean) Norm on $\mathbb{C}^n$

For complex vectors, the **Euclidean norm** is

$$\|x\|_2 = \left(\sum_k |x_k|^2 \right)^{1/2} = \sqrt{x^* x}.$$

(The conjugate transpose handles complex entries.) The **one-norm** is the sum of absolute values, $\|x\|_1 = \sum_k |x_k|$; minimizing it tends to produce **sparse** vectors (most entries zero, a few nonzero). Both are members of the **p -norm (L^p) family**:

$$\|x\|_p = \left(\sum_k |x_k|^p \right)^{1/p}, \quad p \geq 1.$$

- $p = 1$: taxicab / Manhattan norm. - $p = 2$: Euclidean norm. (You could name the $p = 3$ norm yourself — there is no standard name.) - $p \rightarrow \infty$: taking the limit picks out the **maximum absolute element**:

$$\|x\|_\infty = \max_k |x_k|.$$

Worked values for $x = (3, -4)$: $\|x\|_1 = 3 + 4 = 7$, $\|x\|_2 = 5$, $\|x\|_\infty = 4$ (the peak magnitude). **As p increases, the norm decreases** — not a coincidence but a general property.

Why ℓ_∞ Matters: Worst-Case / Minimax

If x is an **error vector**, the ∞ -norm is the **maximum error**. Designing for robustness — minimizing the **worst** error — means minimizing $\|e\|_\infty$, a **minimax** optimization.

Application: FIR Filter Design

A clean application illustrating the three norms is **FIR filter design** in the frequency domain.

Setup

On the (positive) frequency axis, suppose we want a **low-pass filter**: desired response = 1 in the passband and = 0 in the stopband. Sample the frequency axis at f points (e.g., 2000 samples) to get the **desired vector**

$$h_d = (\underbrace{1, 1, \dots, 1}_{\text{passband}}, \underbrace{0, 0, \dots, 0}_{\text{stopband}})^\top \in \mathbb{C}^f.$$

A **realizable** FIR filter has a finite impulse response with r coefficients c ; its frequency response is the Fourier transform of c , written as a matrix product $h = Fc$ (F the $f \times r$ Fourier matrix). The realized response will **not** be perfectly flat — it ripples — so there is an **error vector**

$$e = h_d - Fc.$$

Choice of Norm = Design Criterion

$$\min_c \|h_d - Fc\|_p,$$

and different p give different filters:

- $\ell_2 \rightarrow$ **least squares**. $\min_c \|h_d - Fc\|_2$ has a **closed-form algebraic solution** via the pseudoinverse, $\hat{c} = (F^*F)^{-1}F^*h_d$. Easy.
- $\ell_\infty \rightarrow$ **equiripple / minimax**. $\min_c \|h_d - Fc\|_\infty$ minimizes the worst-case deviation. **No formula** — needs an iterative algorithm. MATLAB's **firpm** (Finite Impulse Response **Parks–McClellan**, “pm”) implements this minimax criterion. (A student asked which MATLAB filter-design command applies; the instructor distinguished special-purpose commands from **firpm**, the general minimax designer.)
- $\ell_1 \rightarrow$ **robust**. $\min_c \|h_d - Fc\|_1$ minimizes total absolute error; no formula, solvable as an LP / via CVX.

The instructor’s point: there is **no universal “best” norm** — it depends on the application and which kind of error matters. “Each norm is easy to solve” only for ℓ_2 (closed form); the others need iteration.

Convexity and Why It Helps

A **convex function** is one where the chord joining any two points on the graph lies **above** the function, and which has **no spurious local minima**. All p -norms ($p \geq 1$) are convex. Minimizing a convex function over a convex search space is **tractable**, even when iterative: tools like **CVX** solve it directly (you state the problem, get the answer), though you don’t get a closed-form formula for ℓ_1/ℓ_∞ . The instructor flags CVX as a tool students will use in the next homework.

Linear-Programming Formulation of ℓ_∞ (and ℓ_1)

ℓ_∞ and ℓ_1 minimization can be cast as **linear programs (LP)** — a special class of convex optimization (an entire industrial-engineering course). The general LP form is

$$\min_x c^\top x \quad \text{s.t.} \quad Ax \leq b \text{ (or } \geq),$$

minimizing the **inner product** $c^\top x$ over a **polyhedral region**.

ℓ_∞ **trick**. Introduce a scalar upper bound t on the peak magnitude. Since $|e_k| \leq t \iff -t \leq e_k \leq t$:

$$\min_{c,t} t \quad \text{s.t.} \quad Fc - h_d \leq t\mathbf{1}, \quad -(Fc - h_d) \leq t\mathbf{1}.$$

Stacking F and $-F$ (with $\mp\mathbf{1}$ columns for t) puts it into the standard LP inequality form, and the objective is the inner product picking out t . At the optimum, t equals the maximum absolute element of the error. So ℓ_∞ **minimization = LP**; a similar trick handles ℓ_1 . You don’t *need* the LP form (CVX’s general convex solver works), but historically people were excited to discover ℓ_1/ℓ_∞ reduce to LP.

For a general p -norm objective such as $p = 3$, the LP trick no longer applies, but the problem is still convex as long as $p \geq 1$. In that case the general convex-optimization route (for example CVX) is the natural tool.

Weighted ℓ_2 -Norms

Beyond the p -norm family, another way to measure size is a **weighted** norm.

Diagonal Weighting

For a diagonal **positive** matrix $D = \text{diag}(d_1, \dots, d_n)$, $d_k > 0$:

$$\|x\|_D = \sqrt{x^* D x} = \left(\sum_k d_k |x_k|^2 \right)^{1/2}.$$

Each coordinate's squared magnitude carries a **weight** d_k . This is useful when not all components are equally significant. **In filter design**, for instance, you weight the **stopband** much more than the passband (e.g., 10,000 vs. 1): on the dB scale a given error in the suppression band makes a “lousy filter,” so you prioritize stopband suppression over passband flatness.

Why the weights must be strictly positive. If some $d_k = 0$, then the nonzero vector e_k has $\|e_k\|_D = 0$, violating definiteness. (Positive **semidefinite** is not enough.)

General Positive-Definite Weighting (Student Q&A)

Replace D by any positive definite matrix W :

$$\|x\|_W = \sqrt{x^* W x}.$$

The instructor asks the class what W does **geometrically** when it is not diagonal. Through guided Q&A: - A student guesses a **correlation** connection — the instructor agrees W is often a **correlation matrix** (in fact the **inverse** correlation appears naturally in applications). - Another student suggests **projecting x onto eigenvector directions** — “Exactly.”

Derivation. Since W is Hermitian PD, it is unitarily diagonalizable: $W = U \Lambda U^*$ with positive eigenvalues. Then

$$x^* W x = x^* U \Lambda U^* x = (U^* x)^* \Lambda (U^* x) = \sum_k \lambda_k |z_k|^2, \quad z = U^* x.$$

Here $z_k = u_k^* x$ is the **projection of x onto eigenvector u_k** (its coordinate in the eigenbasis). So:

Instead of weighting the **coordinate axes** (the diagonal case), a general W weights the **eigenvector directions** of W : you change basis to the eigenvectors $u_1, u_2, \dots$, and the eigenvalues $\lambda_1, \lambda_2, \dots$ become the weights along those (possibly rotated) directions.

Geometrically: drop the projections $z_1 = u_1^* x$, $z_2 = u_2^* x$ onto the eigenvectors; each squared projection is scaled by its eigenvalue.

Weighted ℓ_2 -Norm in Statistics: the Multivariate Gaussian

The PD-weighted 2-norm appears naturally in statistical estimation (the subject of a 530-type course; here just the connection).

Scalar and Independent Gaussians

A single Gaussian has density $\propto \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$ — a bell curve centered at μ , with σ controlling width. For **jointly Gaussian** random variables, the instructor notes a special fact: if they are uncorrelated, then they are independent. That implication is **not true for arbitrary random variables**.

For two independent zero-mean Gaussians with variances σ_1^2 and σ_2^2 , the joint density is the product of marginals:

$$p(x_1, x_2) \propto \frac{1}{2\pi\sigma_1\sigma_2} \exp\left(-\frac{1}{2}\left(\frac{x_1^2}{\sigma_1^2} + \frac{x_2^2}{\sigma_2^2}\right)\right).$$

If the variances are equal, $\sigma_1 = \sigma_2 = \sigma$, this simplifies to

$$p(x_1, x_2) \propto \exp\left(-\frac{x_1^2 + x_2^2}{2\sigma^2}\right) = \exp\left(-\frac{\|x\|_2^2}{2\sigma^2}\right).$$

So for **uncorrelated, equal-variance** Gaussians the ordinary **2-norm squared** appears in the exponent.

Correlated Gaussians → Weighted 2-Norm

When the components are **correlated**, you need the covariance matrix Σ , and the joint density becomes

$$p(x) = \frac{1}{(2\pi)^{n/2} \sqrt{\det \Sigma}} \exp\left(-\frac{1}{2} x^\top \Sigma^{-1} x\right).$$

The exponent is a **weighted 2-norm squared** with weight $W = \Sigma^{-1}$, the **precision (inverse covariance) matrix**. So the PD-weighted norm sits “at the top of the Gaussian,” with the inverse covariance as the weight.

Maximum Likelihood = Weighted Least Squares

Maximizing the Gaussian likelihood means **minimizing** the (negated) exponent $\frac{1}{2} x^\top \Sigma^{-1} x$ — a **weighted least squares** problem. The weighting along eigen-directions interpretation applies: directions of large variance (large eigenvalues of Σ) get small precision weight, so they are penalized less.

L^p Norms for Function Spaces

Norms are not restricted to n -dimensional vectors. For appropriately integrable functions on an interval $[a, b]$, define the L^p **norm**:

$$\|f\|_{L^p} = \left(\int_a^b |f(t)|^p dt \right)^{1/p}.$$

Think of $f(t)$ as a “vector” with **uncountably many** entries (one per t); the integral replaces the sum. This lets us measure **how big a function is** and the **distance between two functions** $\|f - g\|_{L^p}$ — treating each function as a **point** in an infinite-dimensional space.

Polynomial Approximation

Approximate an arbitrary target function $g(t)$ by a polynomial $f(t) = a + bt + ct^2$. The polynomial can only take certain shapes, so it deviates from g at various points (big error here, zero there). To measure performance with a **single number**, define the error function $e(t) = g(t) - f(t)$ and minimize a norm of it:

$$\min_{a,b,c} \|g - f\|_{L^p}.$$

- $p = 2$: minimize total squared error (closed-form via inner products with the polynomial basis). - $p = \infty$: minimize the worst-case pointwise deviation (Chebyshev / equiripple approximation).

The big idea: **norms let us carry two- and three-dimensional geometric intuition (distance, closeness, projection) into high- and infinite-dimensional problems**. The instructor cautions, however, that geometric intuition from 2-D/3-D can sometimes **fail** in high dimensions — “there are surprises in high dimensions.”

Norm Balls (Geometric Understanding)

Before matrix norms, the instructor develops **norm balls**, which give geometric insight — especially into the ℓ_1 -sparsity connection. The **unit ℓ_p -ball** is the region of all vectors with $\|x\|_p \leq 1$ (the boundary is where $\|x\|_p = 1$; “unit” is dropped for brevity).

The ℓ_1 Ball in 2-D (Quadrant-by-Quadrant)

We need $|x_1| + |x_2| \leq 1$. The sign of each term changes by quadrant: - **First quadrant** ($x_1, x_2 > 0$): $x_1 + x_2 \leq 1$. The boundary $x_1 + x_2 = 1$ is a line; $x_1 + x_2 = 0.9, 0.8, \dots$ are parallel lines; the region is the triangle below $x_1 + x_2 = 1$. - **Second quadrant** ($x_1 < 0, x_2 > 0$): $-x_1 + x_2 \leq 1$, another line, another triangle. - Continuing through the third and fourth quadrants gives four triangular pieces that assemble into a **diamond** (a rotated square) with corners at $(\pm 1, 0)$ and $(0, \pm 1)$.

So the ℓ_1 ball is a **diamond** — “this is a new geometry; your ball is no longer the round thing.”

The ℓ_2 and ℓ_∞ Balls in 2-D

- ℓ_2 : $x_1^2 + x_2^2 \leq 1$ is the familiar **circular** disk.
- ℓ_∞ : $\max(|x_1|, |x_2|) \leq 1$ means **both** $|x_1| \leq 1$ and $|x_2| \leq 1$ — a **square** (axis-aligned, **not** tilted) with corners at $(\pm 1, \pm 1)$. On the right edge $x_1 = 1$ (with $|x_2| \leq 1$), the peak value is 1; inside, both coordinates are < 1 .

Nesting Comparison

Overlaying the three: the ℓ_1 **diamond is inside the ℓ_2 circle, which is inside the ℓ_∞ square**. This visualizes $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$: - A point with $\|x\|_1 > 1$ but $\|x\|_2 \leq 1$ sits **outside** the diamond but **inside** the circle (and square). - A point with $\|x\|_1, \|x\|_2 > 1$ but $\|x\|_\infty \leq 1$ sits outside both the diamond and circle but inside the square.

The 3-D Balls and “Orthant” Terminology

In $\mathbb{R}^3$: - ℓ_2 **ball**: $x_1^2 + x_2^2 + x_3^2 \leq 1$ — a **sphere** (solid spherical region) around the origin. - ℓ_∞ **ball**: $\max(|x_1|, |x_2|, |x_3|) \leq 1$ — a **cube**. - ℓ_1 **ball**: $|x_1| + |x_2| + |x_3| \leq 1$ — an **octahedron**.

For the ℓ_1 ball, in the all-positive region the boundary $x_1 + x_2 + x_3 = 1$ is a **hyperplane** — note $x_1 + x_2 + x_3 = \langle x, \mathbf{1} \rangle = 1$, a plane orthogonal to $\mathbf{1}$, shifted off the origin (since the right side is 1). Each of the 2^n sign regions contributes a triangular face, giving the octahedron.

Terminology Q&A. Students try to name the n -dimensional generalization of “quadrant.” In 2-D the axes divide the plane into 4 regions (quadrants); in 3-D into 8 regions; in n -D into 2^n regions. The general name (after some back-and-forth) is **orthant** — each orthant being where all components have fixed signs.

Application: Overdetermined Systems and Norms

One of the most useful parts of the course: using norms to solve linear systems.

Overdetermined (Tall) Systems

An **overdetermined** system has **more equations than unknowns** — A is a **tall** matrix. The potential problem: the column space (range) of A does **not** cover the whole target space, so if b is **not** in the range of A there is **no solution** (the system is **inconsistent**). A “lazy researcher” stops there — but we don’t.

Closest Point in the Range

Instead, find the point $\hat{b} = Ax^*$ in the range of A that is **as close as possible to b** , minimizing the **norm** of the error $b - Ax$. But “close” requires a choice of norm:

$$\min_x \|b - Ax\|.$$

The ℓ_2 Case Has a Formula

If we use the **2-norm** (and A is full rank), the solution is the **orthogonal projection** of b onto the range of A with respect to the Euclidean inner product. Recalling the projection formula for a non-orthonormal basis,

$$\hat{b} = A(A^*A)^{-1}A^*b,$$

and the multiplier of A is the solution

$$x^* = (A^*A)^{-1}A^*b.$$

This closed-form formula is **why people gravitate to the 2-norm**: no iterative algorithm needed. (For ℓ_1/ℓ_∞ there is no such formula — use CVX or LP.)

For ℓ_1 and ℓ_∞ , the same LP reductions discussed earlier can be used; for a general p -norm residual, CVX can still solve the convex problem when $p \geq 1$, but it will not usually be a linear program.

Why the 2-Norm Has a Formula but ℓ_1/ℓ_∞ Don't

The 2-norm is **associated with an inner product**: $\langle x, y \rangle = y^\top x$ (real case), and $\|x\|_2 = \sqrt{\langle x, x \rangle}$. That inner-product structure is what yields the orthogonal-projection solution. The ℓ_1 and ℓ_∞ norms have **no associated inner product** — you cannot write them as $\sqrt{\langle x, x \rangle}$ — so the projection machinery does not apply, and you must optimize numerically. (Inner-product norms will be developed in the inner-product lectures.)

Statistical Justification: Gaussian Noise $\rightarrow$ Least Squares

The 2-norm criterion is also **statistically optimal** under a Gaussian-noise model. Suppose the measurement is

$$b = Ax + n,$$

with n Gaussian noise. If n is **white** (covariance = $\sigma^2 I$), the **maximum-likelihood** estimate minimizes $\|b - Ax\|_2^2$ — exactly **least squares** (the σ^2 scales out). This is why minimizing the sum of squared errors is “justified”: b was really in the range, and only Gaussian noise pushed it out.

Correlated Noise $\rightarrow$ Weighted Least Squares

If the noise is **correlated** with covariance Σ , the ML estimate uses the **weighted** norm with $W = \Sigma^{-1}$:

$$\min_x \|b - Ax\|_{\Sigma^{-1}}^2 = \min_x (b - Ax)^\top \Sigma^{-1} (b - Ax).$$

For uncorrelated noise $\Sigma = \sigma^2 I$ this collapses back to ordinary least squares because the scalar σ^2 only rescales the objective. With known noise statistics you insert Σ^{-1} as the weight; this again weights the eigen-directions of Σ by the inverse eigenvalues. (For non-full-rank A , additional tricks are needed — deferred to later lectures.)

Instructor Remarks and Study Guidance

- **No universally best norm** — the norm encodes which error you care about: ℓ_1 (total absolute, robust, sparsity-promoting), ℓ_2 (total squared, smooth, closed-form), ℓ_∞ (worst-case, equiripple, minimax).
- **As p grows the norm shrinks**: $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$. Norm balls: ℓ_1 **diamond** $\subset \ell_2$ **circle** $\subset \ell_\infty$ **square** (sphere/cube/octahedron in 3-D). The n -D sign regions are **orthants** (2^n of them).
- **Convexity** ($p \geq 1$) makes all these minimizations tractable; ℓ_1 and ℓ_∞ further reduce to **linear programs**; use **CVX** for general convex p , including cases such as $p = 3$ that are not LPs. CVX is explicitly tied to the next homework.
- **Weighted ℓ_2** with PD weight W weights the **eigenvector directions** of W by its eigenvalues. It is the natural norm for **multivariate Gaussian** estimation, with weight = precision matrix Σ^{-1} .

- **The 2-norm is special** because it comes from an **inner product**, giving the closed-form least-squares / orthogonal-projection solution $x^* = (A^*A)^{-1}A^*b$. ℓ_1 and ℓ_∞ have no associated inner product.
- **Least squares is ML estimation** under white Gaussian noise; correlated noise gives **weighted** least squares with Σ^{-1} .

Source and Coverage Note

Source: `corrected/lecture19_corrected.md`.

Coverage: Recap of norm axioms and the p -norm family ($\ell_1/\ell_2/\ell_\infty$, worked $(3, -4)$ values, monotonicity in p); ℓ_∞ as worst-case/minimax; FIR filter design (low-pass desired vector, $h = Fc$, error vector, three criteria — ℓ_2 least-squares closed form, ℓ_∞ `firpm`/Parks–McClellan equiripple, ℓ_1 robust — with the filter-command Q&A); convex-function definition; LP formulation of ℓ_∞ (scalar bound t , polyhedral region, inner-product objective) and the ℓ_1 analog; weighted ℓ_2 -norm (diagonal weighting with filter-design stopband example, strict-positivity requirement, general PD W with the correlation/eigenvector student Q&A and eigen-decomposition derivation); multivariate Gaussian (scalar/independent $\rightarrow$ 2-norm, correlated $\rightarrow$ weighted 2-norm with precision matrix, MLE = weighted least squares); L^p function-space norms and polynomial approximation; detailed norm-ball geometry (quadrant-by-quadrant ℓ_1 diamond, ℓ_2 circle, ℓ_∞ square, nesting comparison, 3-D sphere/cube/octahedron, hyperplane faces, orthant terminology Q&A); overdetermined systems and norms (tall/inconsistent, closest point, ℓ_2 projection formula $x^* = (A^*A)^{-1}A^*b$, 2-norm $\leftrightarrow$ inner-product connection and why ℓ_1/ℓ_∞ lack one, Gaussian-noise/MLE justification, correlated-noise weighted least squares).